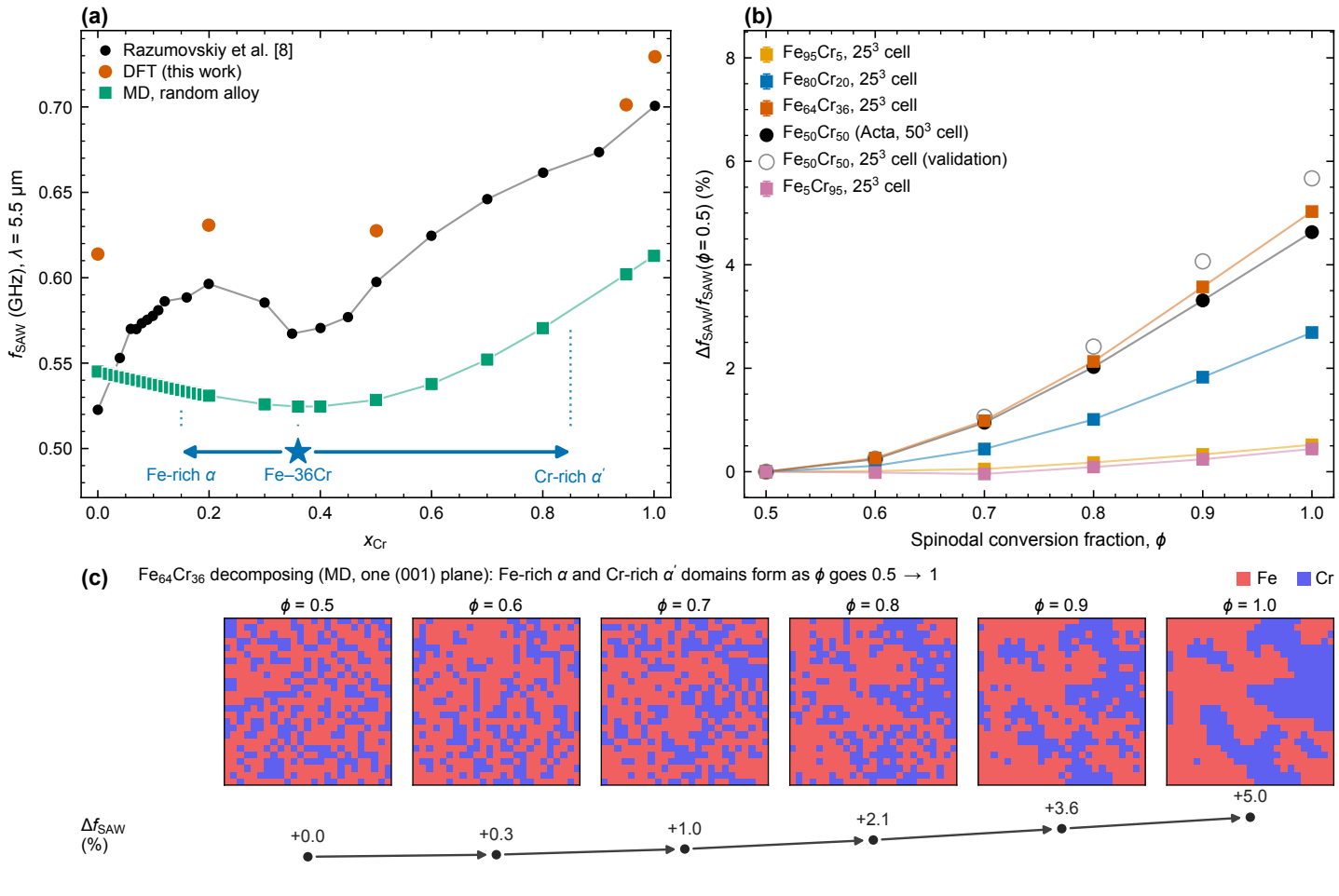

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
f_{SAW} (GHz), λ = 5.5 μm
0.70
0.65
0.60
0.55
0.50
0.0
0.2
0.4
0.6
0.8
1.0
x_{Cr}
Fe-rich α
Fe–36Cr
Cr-rich α'
(b)
$Fe_{95}Cr_5$, 25^3 cell
$Fe_{80}Cr_{20}$, 25^3 cell
$Fe_{64}Cr_{36}$, 25^3 cell
$Fe_{50}Cr_{50}$ (Acta, 50^3 cell)
$Fe_{50}Cr_{50}$, 25^3 cell (validation)
Fe_5Cr_{95}, 25^3 cell
$\Delta f_{SAW}/f_{SAW}(\phi = 0.5)$ (%)
8
6
4
2
0
0.5
0.6
0.7
0.8
0.9
1.0
Spinodal conversion fraction, ϕ
(c)
$Fe_{64}Cr_{36}$ decomposing (MD, one (001) plane): Fe-rich α and Cr-rich α' domains form as ϕ goes 0.5 → 1
Fe
Cr
ϕ = 0.5
ϕ = 0.6
ϕ = 0.7
ϕ = 0.8
ϕ = 0.9
ϕ = 1.0
Δf_{SAW} (%)
+0.0
+0.3
+1.0
+2.1
+3.6
+5.0